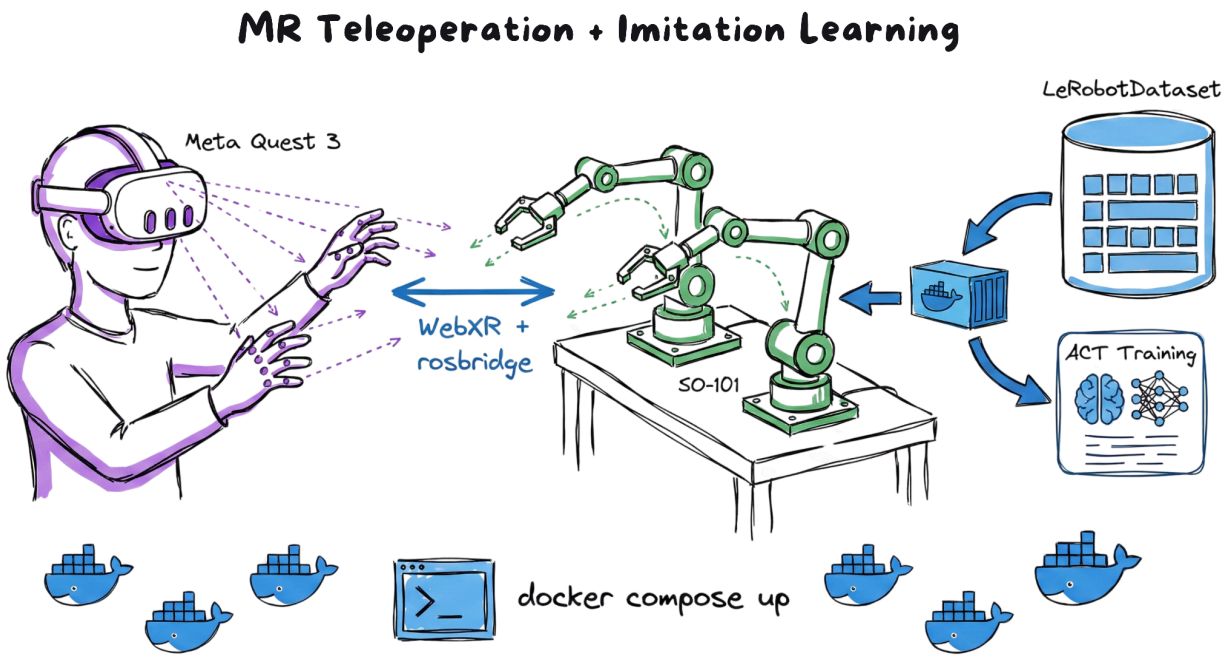
MR Teleoperation of SO-101 Robotic Arm for Imitation Learning

DevOps for Cyber-Physical Systems | University of Bern



Context

Imitation learning for robotic manipulation depends entirely on the quality and scale of demonstration data. The standard approach uses a physical leader-follower arm setup: the operator moves the leader arm, the follower arm mimics the motion, and both trajectories are recorded. This works, but it physically tethers the operator to the robot and limits the workspace to wherever the leader arm can reach.

Mixed reality offers a different interface. Instead of puppeteering a physical leader arm, the operator wears a Meta Quest 3 headset and sees the real robot arms through passthrough while virtual overlays show target poses, trajectory previews, and recording status. Hand tracking or controllers command the robot directly in the operator's physical space. This decouples the operator from the leader arm hardware and opens up possibilities like ergonomic improvements and richer spatial awareness during demonstrations.

Motivation

The SO-101 from HuggingFace LeRobot is a 6-DOF open-source robotic arm designed specifically for imitation learning. It ships as a leader-follower pair with support for ACT (Action Chunking with Transformers) training through the LeRobot framework. The hardware is cheap ($220-$240 per kit), well-documented, and used widely in robotics hackathons and research labs.

But the leader arm approach has real limitations. The operator must sit right next to the robot. Workspace is constrained by the leader arm's reach. And there is no depth perception beyond what your eyes give you from one angle. Open-TeleVision (CoRL 2024) showed that headset-based teleoperation with active stereoscopic feedback produces higher-quality demonstrations. Nobody has applied this idea to the SO-101 ecosystem with MR passthrough and a proper containerized data pipeline.

Goal

The student will build an MR teleoperation system for two SO-101 robotic arms using Meta Quest 3, integrated with the LeRobot framework for imitation learning. The full data collection pipeline runs on containerized infrastructure. The system includes:

MR Teleoperation Interface: WebXR-based application extending SimNav-XR that maps Meta Quest 3 controller poses (or hand tracking) to SO-101 joint commands. The operator sees the real workspace through MR passthrough while virtual overlays show target poses, trajectory previews, joint limits, and recording status on top of the physical arms.

WebSocket Bridge: rosbridge_suite exposing ROS2 topics over WebSocket to the browser-based XR interface. Joint commands flow from WebXR to robot at 30Hz minimum. Camera feeds, joint states, and gripper positions stream back to the headset for real-time feedback via roslibjs.

Containerized Data Pipeline: Docker containers for: (1) SO-101 motor control and sensor reading, (2) camera capture and synchronization, (3) demonstration recording in LeRobotDataset format, (4) data validation and quality checks. Docker Compose orchestrates the full stack with a single command.

LeRobot Integration: Recorded MR demonstrations are stored in LeRobotDataset format compatible with HuggingFace Hub. A containerized training job can run ACT policy training on collected data. CI pipeline validates dataset integrity and runs basic training sanity checks on every new batch of demonstrations.

Benchmarking and Comparison: Quantitative comparison of MR teleoperation vs. physical leader arm teleoperation on the same manipulation tasks (pick-and-place, stacking). Metrics include task success rate, trajectory smoothness, data collection speed (demonstrations per hour), and resulting ACT policy performance.

Requirements

Comfortable with Python. Some experience with web development (JavaScript, Three.js) is helpful but can be learned. Willingness to work with real hardware (two SO-101 arms, Meta Quest 3 headset, USB cameras). Interest in human-robot interaction and machine learning for robotics. No prior MR development experience needed, as SimNav-XR provides the foundation to build on.

Pointers

• SimNav-XR (Aryan et al., 2026) — Extended reality platform for mobile robot simulation using ROS2 and Unity3D

• HuggingFace LeRobot — Open-source framework for robot learning with imitation learning, RL, and VLA models

• SO-101 Quickstart Guide — Official setup, calibration, and data collection guide for the SO-101 arms

• Open-TeleVision (CoRL 2024) — Cheng et al. on headset-based teleoperation with immersive active visual feedback for imitation learning

• Open-TeleVision Source Code — Implementation of headset-based teleoperation with ACT training pipeline

• unity_ros_teleoperation (ETH Zurich) — Unity XR teleoperation with ROS integration for Quest 3, OpenXR, and hand tracking

• VR Dual-Arm Teleoperation (MDPI 2026) — Framework for dual-arm manipulation integrating simulation and real-world control via headset

• roslibjs — JavaScript library for interacting with ROS2 via rosbridge WebSocket from browser-based XR apps

• ACT: Action Chunking with Transformers — Reference implementation of the ACT algorithm used with ALOHA and LeRobot

• SO-ARM101 Product Page (Hiwonder) — Hardware specs, dual-camera system, and LeRobot framework integration details

• Docker Compose Documentation — Multi-container orchestration for the data collection pipeline

• Meta Quest Developer Hub — WebXR development guides for Meta Quest 3, hand tracking, and passthrough MR

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